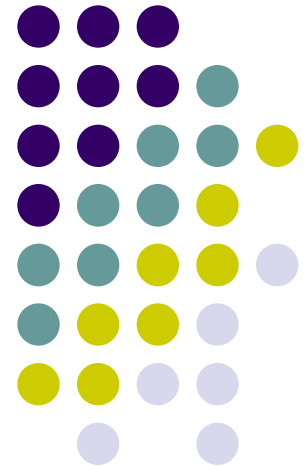


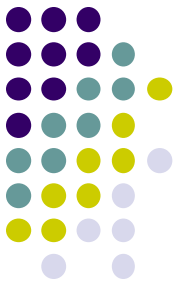
Domain models as basis for intelligent applications

Lecture 7

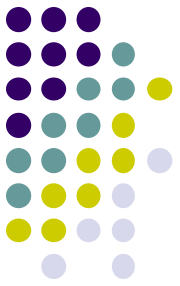
Dynamic Modelling for Human-centered Systems



Course outline



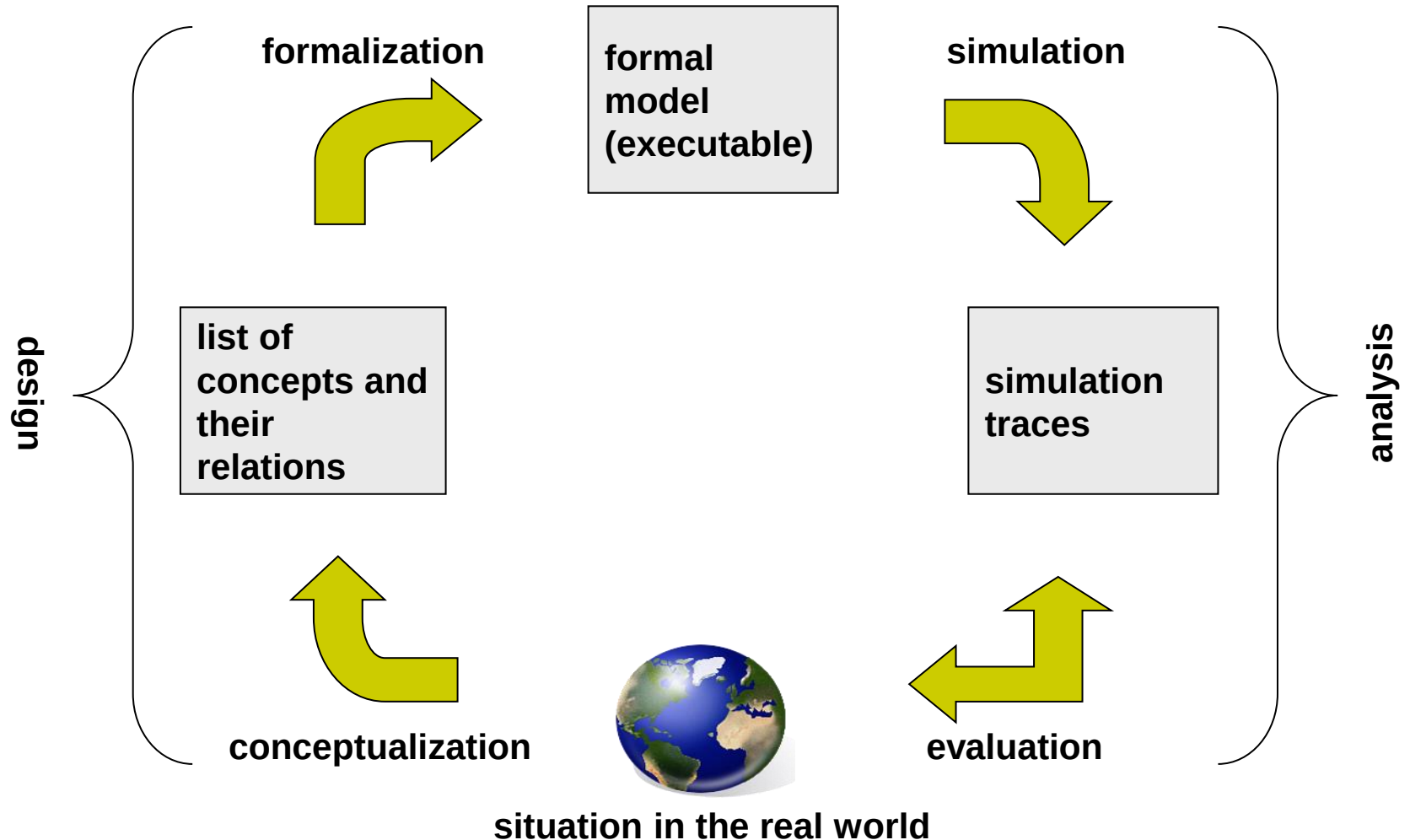
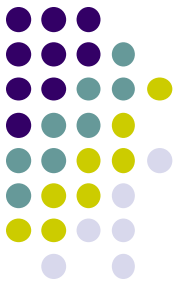
- Week 1: what is dynamic modelling?
 - Week 2: modelling social processes
 - Week 3: modelling behaviour
 - **Week 4: domain models as basis for intelligent applications**
 - Week 5: agent-based modelling
 - Week 6: analysis and support models
 - Week 7: wrap-up
- } domain models

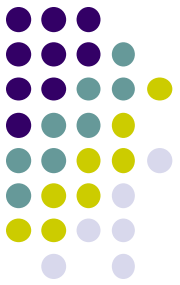


Overview of Today's Lecture

1. Creating a domain model
 - Conceptualisation (concepts and relations)
 - Formalisation (ontology and LEADSTO)
2. How to use models as basis for intelligent systems?

Modelling & Simulation Cycle

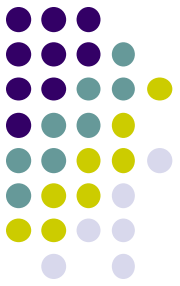




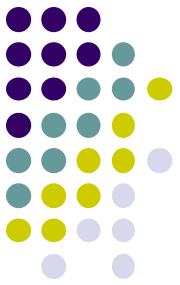
Conceptualisation - Static

- Which concepts play a role in a given description?
- Result: list of concepts
- Useful distinction:
 - Input states
 - Internal states
 - (sensory representation, preparation, ...)
 - Output states
 - External states

Video - Paro Robot



<https://www.youtube.com/v/2ZUn9qtG8ow>



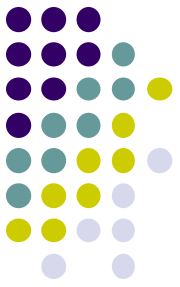
Assignment

Conceptualisation (static)

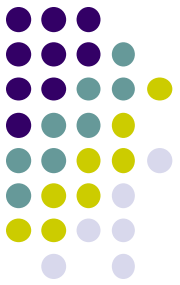
- List the relevant concepts given in this text
- Distinguish input, internal, and output states (external states can be ignored)

- with your neighbour

Case Study - Paro

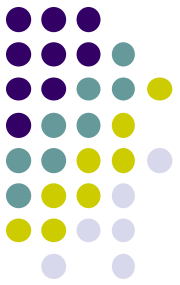


“Paro is a robot seal, used for therapeutic purposes. By moving a switch on Paro’s back to the left or the right, she can be set to two modes, namely a happy and a sad mode, respectively. In the happy mode, Paro produces a loud sound when touched by the user. In the sad mode, Paro produces a soft sound when touched by the user. Moreover, in the happy mode, Paro can react to voices: when hearing a human voice, she moves her tail and blinks her eyes.”



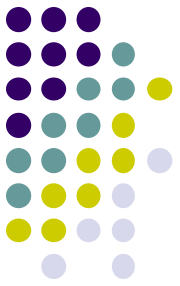
List of Concepts

- Input states (observations)
 - ...
 - ...
 - ...
- Internal states
 - ...
 - ...
 - ...
- Output states (actions)
 - ...
 - ...
 - ...



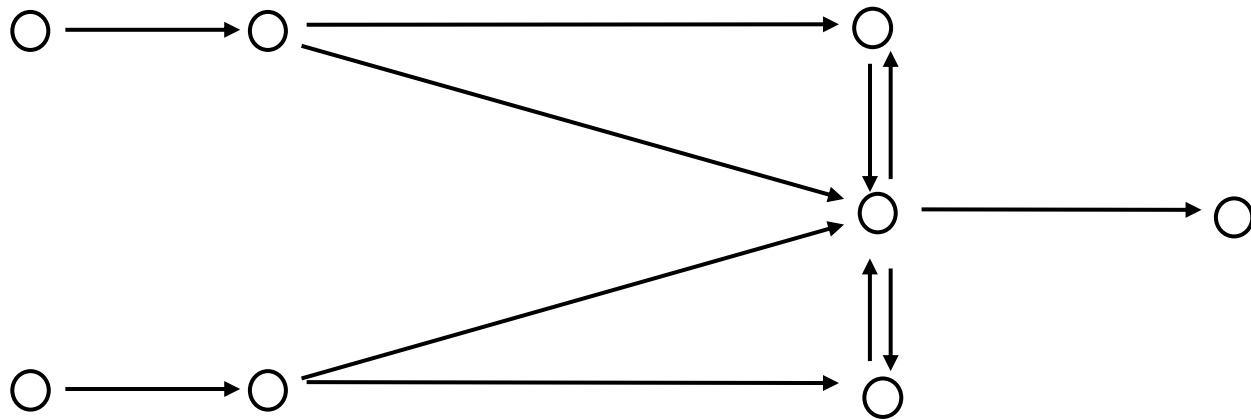
List of Concepts

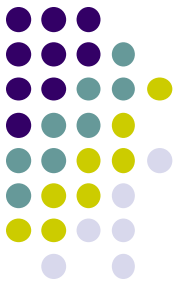
- Input states
 - Switch (left, right)
 - Touch
 - Voice
- Internal states
 - Modes (happy, sad)
 - Sensory representations (of input states, e.g. sr_voice)
 - Preparation states (for actions, eg ps_sound)
- Output states (actions)
 - Sounds (loud, soft)
 - Tail movement
 - Eye blinks



Conceptualisation - Dynamic

- Which concepts influence each other?
- Result: picture with circles and arrows





How to draw diagrams?

- <https://mermaid.live/>

stateDiagram-v2

direction LR

Ext.Concept1 --> SR1

Ext.Concept1 --> SR2

state agent {

SR2 --> Internal

SR1 --> Internal

}

Internal --> Action

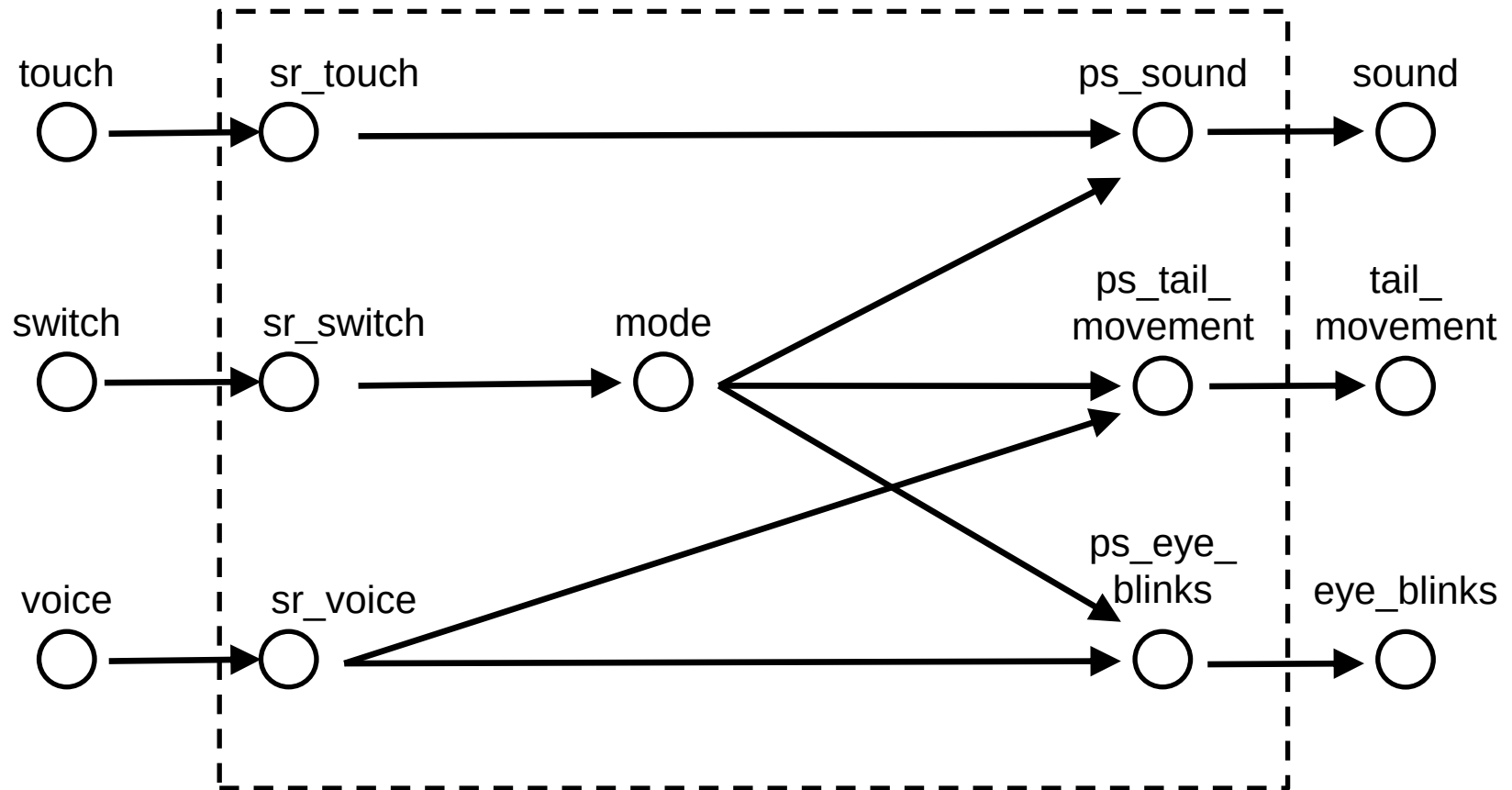
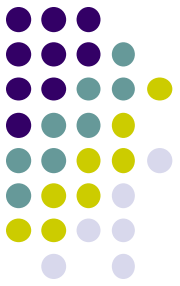


Assignment

Conceptualisation (dynamic)

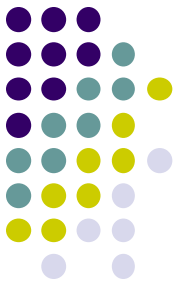
- Identify relations between these concepts
- Draw these in a picture
(details of relations can be ignored)
- First group to show a correct picture wins
 - come to the front

Relations between Concepts



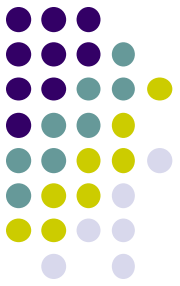
Formalisation

LEADSTO Language

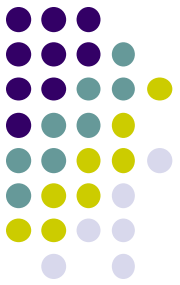


- Basic idea:
 - States of the world are described in terms of logical predicates
 - = ontology with predicates and sorts
 - Dynamics of the world are expressed as temporal if-then rules
- Important distinction between:
 - LEADSTO language and
 - L2-Python simulation environment

LEADSTO - example ontology

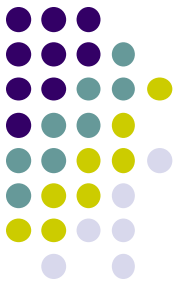


<i>concept</i>	<i>predicate</i>	<i>example use</i>
it is raining	raining	raining
the current time is X o'clock	current_time: INTEGER	current_time(15) current_time(X)
agent A observes X	observed : AGENT x INFO_ELEMENT	observed(animal, no_screen) observed(A, X)
agent A performs action Y	performed : AGENT x ACTION	performed(animal, goto(p2)) performed(A, Y)
...		



LEADSTO - example sorts

<i>sort</i>	<i>description of use</i>	<i>elements</i>
AGENT	ambient and human agents	john, mary, animal, agent1, agent2, ...
INFO_ELEMENT	information elements	no_screen, at_position(F:FOOD, P:POSITION), ...
FOOD	types of food	food, icecream, cheese, ...
POSITION	positions	p1, p2, ...
ACTION	actions	eat, goto(P:POSITION), ...
REAL	the real numbers	0, 0.01, -3.5, 1000, ...
...		

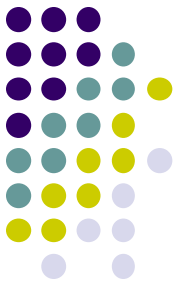


Assignment

Formalisation (static)

- Formalise the list of concepts
- Give all predicates and sorts

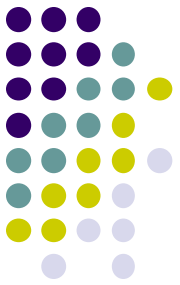
Formal Ontology



<i>Concept</i>	<i>Predicate</i>
observations (switch, touch, voice)	observed : AGENT x INFO_ELEMENT
internal states (mode, sr, ps)	...
actions (sounds, tail, eye blinks)	...

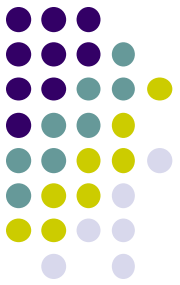
<i>Sort</i>	<i>Elements</i>
AGENT	paro
INFO_ELEMENT	...
...	...
...	...
...	...
...	...

Formal Ontology



<i>Concept</i>	<i>Predicate</i>
observations (switch, touch, voice)	observed : AGENT x INFO_ELEMENT
internal states (mode, sr, ps)	mode : AGENT x MODE sr: AGENT x INFO_ELEMENT ps : AGENT x ACTION
actions (sounds, tail, eye blinks)	performed : AGENT x ACTION

<i>Sort</i>	<i>Elements</i>
AGENT	paro
INFO_ELEMENT	switch(DIRECTION), touch, voice
ACTION	make_sound(SOUND), move_tail, blink_eyes
DIRECTION	left, right
MODE	happy, sad
SOUND	loud, soft



LEADSTO - Example Rules

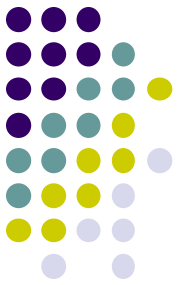
at any point in time,
if the animal observes that **food** is present at p2
 and it observes that **no screen** is present,
then it will **go to** position p2

observed(animal, at_position(food, p2)) $\wedge$

observed(animal, no_screen)



performed(animal, goto(p2))



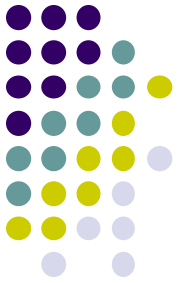
Assignment

Formalisation (dynamic)

- Formalise all dynamic relations
- Use the formal LEADSTO notation

observed(A, at_position(food, p2)) & no_screen →
performed(A, performed(animal, goto(p2)))

Formal Rules



R1a

...

R1b

...

...

R2a

...

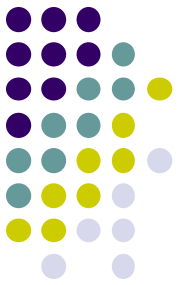
R2b

...

R3

...

...



Formal Rules

R1a

$\text{observed}(\text{A:AGENT}, \text{I:INFO_ELEMENT}) \rightarrow \text{sr}(\text{A}, \text{I})$

R1b

$\text{sr}(\text{A:AGENT}, \text{switch}(\text{right})) \rightarrow \text{mode}(\text{A}, \text{happy})$

$\text{sr}(\text{A:AGENT}, \text{switch}(\text{left})) \rightarrow \text{mode}(\text{A}, \text{sad})$

R2a

$\text{mode}(\text{A:AGENT}, \text{happy}) \ \& \ \text{sr}(\text{A}, \text{touch}) \rightarrow \text{ps}(\text{A}, \text{make_sound}(\text{loud}))$

R2b

$\text{mode}(\text{A:AGENT}, \text{sad}) \ \& \ \text{sr}(\text{A}, \text{touch}) \rightarrow \text{ps}(\text{A}, \text{make_sound}(\text{soft}))$

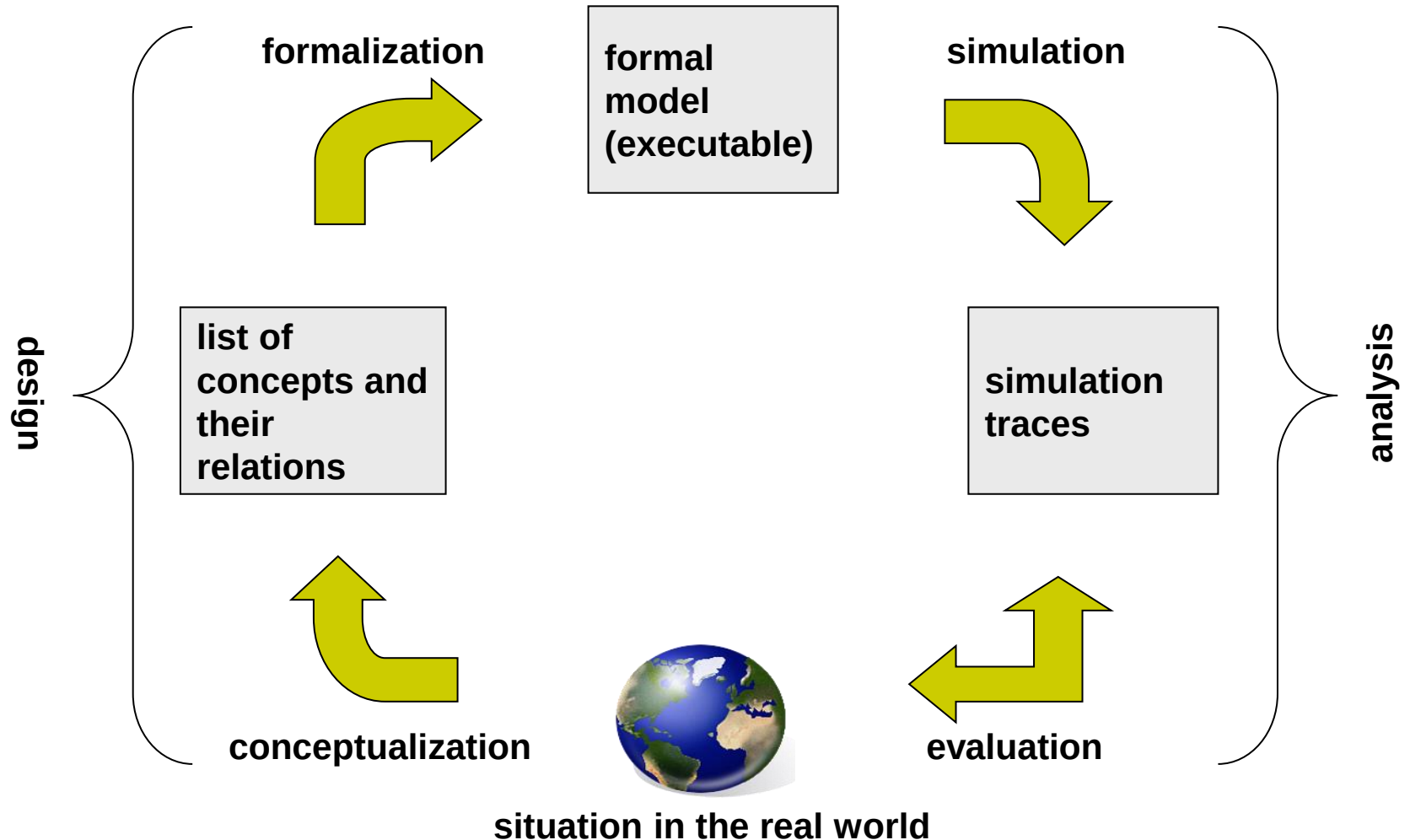
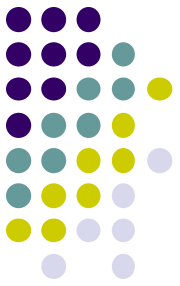
R3

$\text{mode}(\text{A:AGENT}, \text{happy}) \ \& \ \text{sr}(\text{A}, \text{voice}) \rightarrow \text{ps}(\text{A}, \text{move_tail})$

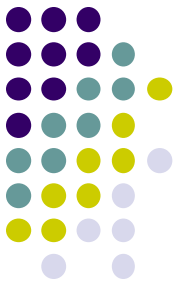
$\text{mode}(\text{A:AGENT}, \text{happy}) \ \& \ \text{sr}(\text{A}, \text{voice}) \rightarrow \text{ps}(\text{A}, \text{blink_eyes})$

$\text{ps}(\text{A:AGENT}, \text{Ac:ACTION}) \rightarrow \text{performed}(\text{A}, \text{Ac})$

Modelling & Simulation Cycle

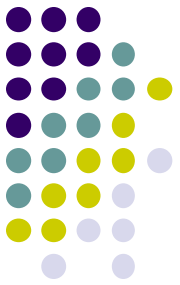


Discussion



- Modelling: an art, rather than a science?
- Are there any guidelines at all?

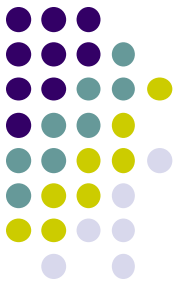




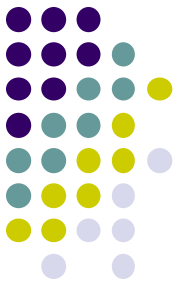
Modelling - considerations

- Important considerations when designing a model:
 - Why do I build a model at all?
 - Simple is good (principle of parsimony)
 - Use concepts that can be made concrete
 - Think about level of granularity (physiological, cognitive, social, ...)
 - Distinguish input, internal, output, external
 - Formalisation: qualitative or quantitative?
 - Rules: type of influence (positive, negative, conjunction)?

Dynamic models as basis for intelligent applications



- What do we mean with intelligent applications?
- How can we use dynamic models for making computer systems intelligent?



Ambient Intelligence

- Video: the vision of **Ambient Intelligence**
- <https://www.youtube.com/watch?v=1M3EYGZ801U>

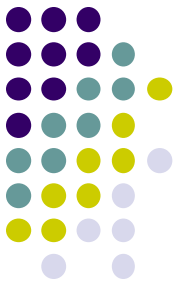


Ambient Intelligence



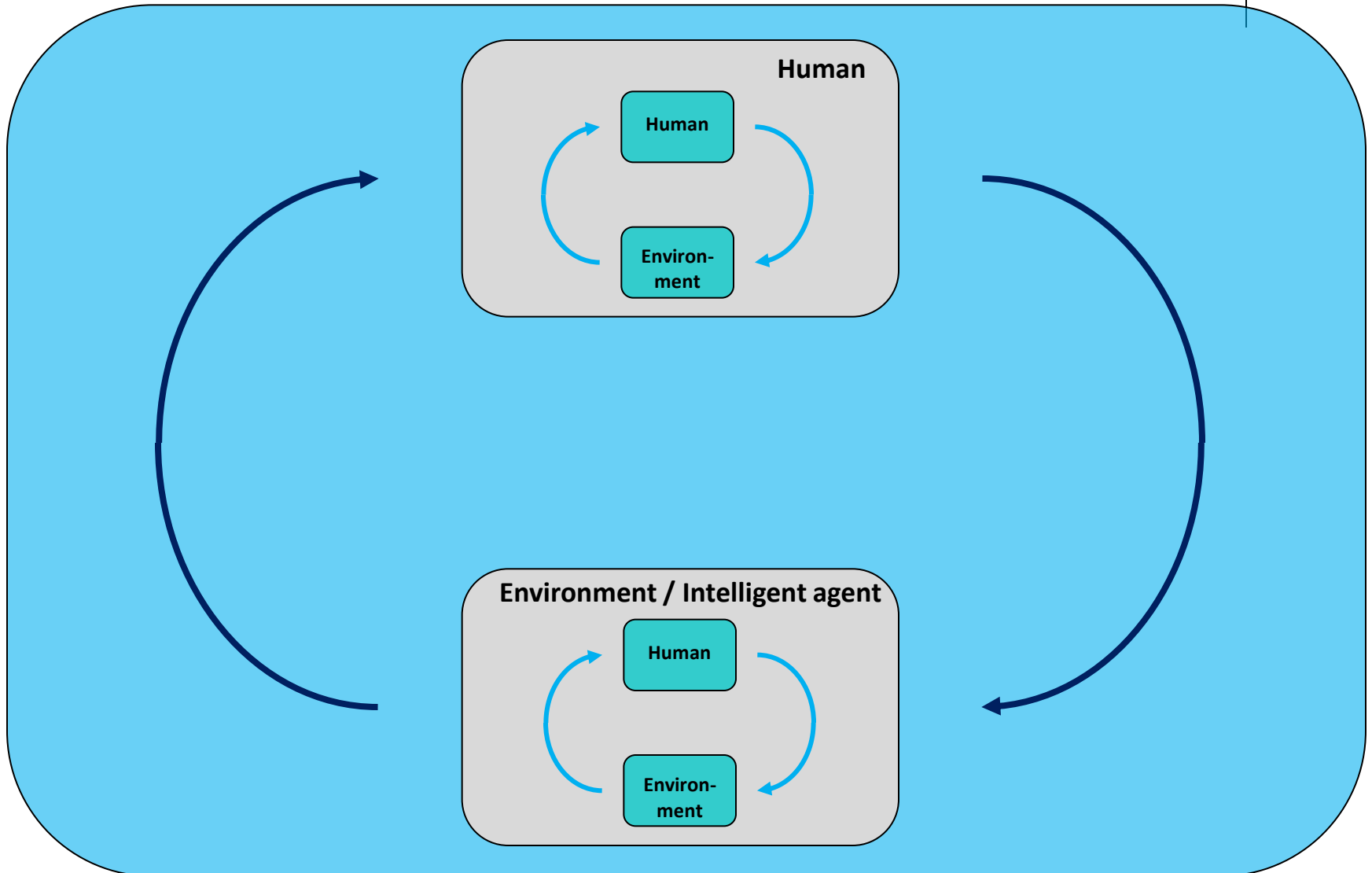
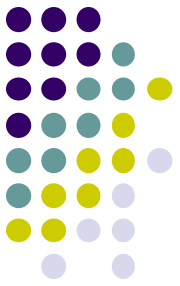
Hanover Helps: Hello, how can I help you?

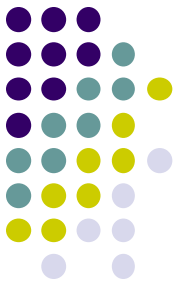
Building ambient intelligent systems



- Ambient intelligent systems:
 - interact and cooperate with users
 - adapt to needs and preferences of users
- What is needed?
 - **knowledge** about humans and how they function
- In this course:
 - incorporate **dynamic models** about human functioning
 - focus on conceptual design, not on implementation

Ambient Intelligence: Global Idea

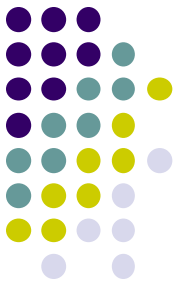




Dynamic model

- A model:
 - representation of an object, phenomenon, etc.
 - can be used to simulate this process
⇒ *dynamic computer model*

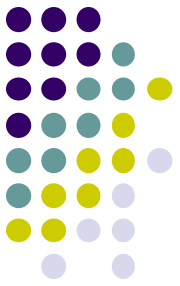
Integrating models in Applications



By incorporating a model within an application, a smart system can be built that has **knowledge** about human functioning, and is able to **reason** about this (including effects over **time**)

Model-based reasoning

Users of models

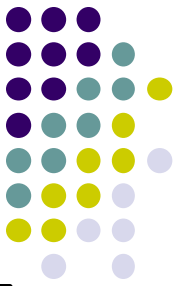


- So far:
 - researchers
 - policy makers



- Now:
 - applications

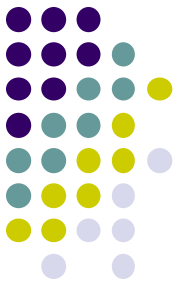




Question

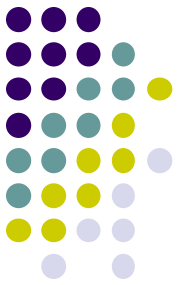
- If we reason about humans using dynamic models within Ambient Intelligent applications, **what** type of conclusions do we want to draw?

Think about this for 1 minute



Two types of reasoning

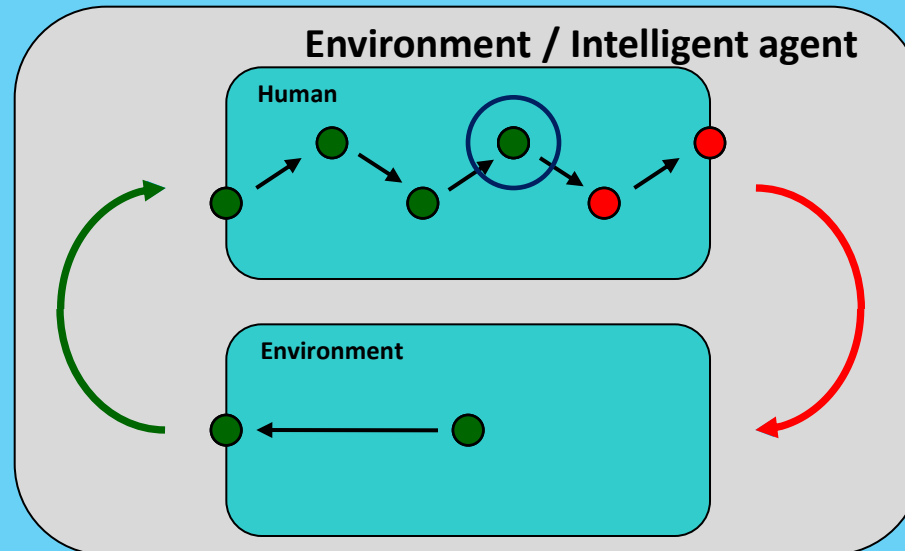
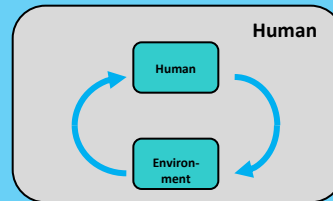
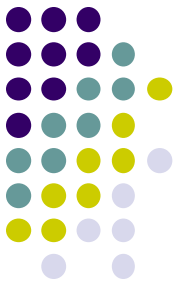
- What is going on?
 - making **assessments** of the current situation
- What will happen if?
 - **predicting** the effect of possible interventions



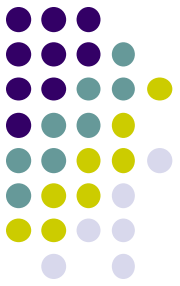
Two types of reasoning

- Making **assessments** of the current situation
 - ‘This user has a sad face and has used negative words over the last 2 hours, he is probably depressed’
 - ‘This driver is blinking with her eyes all the time, she is probably getting tired’
- **Predicting** the effect of possible interventions
 - ‘If the user goes for a walk in the park, he might feel better’
 - ‘If the driver takes a break of 30 minutes, she will recover’

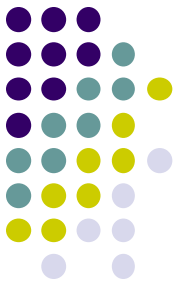
Using models within intelligent applications



Integrating models within intelligent applications



- Examples of **domain models** of human functioning:
 - **physiological** processes
(circulatory system, medicine uptake, exhaustion, ...)
 - **cognitive/behavioural** processes:
(decision process, emotions, beliefs, ...)
 - **social** processes:
(spread of information between people, group behaviour...)



Next lecture

- Principle of agent-based design
- Ontology for agent-based models